

Planogram Recommendations — Cooler CLR-1029

Executive conclusion

I recommend three actions for the current layout, with one of them explicitly gated by a data-quality check:

1. **Widen P2 — Energy Drink B — at shelf 1, cell 2**, from one facing to two if the physical shelf can accommodate the additional 8 cm.
2. **Shrink P4 — Chips Family Pack — at shelf 1, cell 4**, from two facings to one, releasing 8 cm for a stronger seller or another test item.
3. **Investigate P6 — Kombucha — at shelf 2, cell 2 before deciding whether to retain it, shrink it, or replace it.** The observed paid demand is very weak and the return rate is exceptionally high, but the signal may reflect a product/recognition or operating problem rather than genuine shopper demand.

I would **not make a placement decision for shelf 2, cell 3 (P7 — Mystery Wrap) based solely on the recognition data**. That slot has 49 consumer TAKE events attributed to UNKNOWN , all flagged as misplaced and with a mean recognition probability of 0.393. Transactions nevertheless contain 26 paid P7 sales, and the recognized P7 TAKE count is also 26. This is a material attribution anomaly that should be resolved before using the slot as a clean demand signal.

Data and investigation approach

The analysis used all three supplied files for the 45-day period. I first checked file sizes, schemas, date ranges, and key categorical values. I then treated transactions.csv as the billing ground truth for paid demand, while using recognition_events.csv to examine shopper takes/returns, recognition confidence, misplacement flags, and restocking activity. Stocker PLACE events were kept separate from consumer activity so replenishment would not be mistaken for demand.

The main investigation steps were:

Step	Operation	Purpose
1	Read the planogram and map each (shelf_ordinal, cell_ordinal) to its current product	Establish exact current slots, widths, and facing counts

2	Group transactions by product and type	Calculate SALE units/revenue, ZERO units, and return rate
3	Group consumer recognition events by product and direction	Cross-check TAKE/RETURN activity against billing
4	Group consumer recognition by slot and compare attributed product with the planogram map	Detect products being recognized in the wrong cell or as UNKNOWN
5	Separate stocker PLACE events	Check replenishment cadence without treating it as shopper demand
6	Inspect low-confidence and misplace-flagged events	Identify signals that should not drive an immediate layout change

The accompanying workbook, [planogram_investigation.xlsx](#), contains the validation table, planogram, product performance pivot, UFO summary, product-level recognition pivot, slot activity, slot-mismatch matrix, stock placements, cart transaction summary, and daily product activity tabs. The analysis script is also included so the calculations can be rerun.

Evidence from the data

There were **1,206 transaction rows**: 922 SALE lines totaling **922 paid units and \$2,820.50 revenue**, and 284 ZERO lines totaling 284 picked-up-and-returned units. There were 1,611 recognition events: 1,539 consumer TAKE/RETURN events and 72 stocker PLACE events. The event and transaction timestamps cover June 1 through July 15, 2026, consistent with a 45-day observation window.

The product-level performance is:

Current slot	Product	Width / facings	SALE units	Revenue	ZERO units	Return or pickup
Shelf 1, cell 2	P2 — Energy Drink B	8 cm / 1	506	\$1,619.20	0	0.0%
Shelf 2, cell 1	P5 — Sparkling Water	8 cm / 1	107	\$214.00	7	6.1%

Shelf 1, cell 3	P3 — Protein Bar A	6 cm / 1	103	\$257.50	5	4.6%
Shelf 1, cell 1	P1 — Energy Drink A	8 cm / 1	87	\$278.40	4	4.4%
Shelf 2, cell 4	P8 — Yogurt Drink	8 cm / 1	73	\$204.40	13	15.1%
Shelf 2, cell 3	P7 — Mystery Wrap	8 cm / 1	26	\$156.00	0	0.0%
Shelf 2, cell 2	P6 — Kombucha	8 cm / 1	18	\$81.00	255	93.4%
Shelf 1, cell 4	P4 — Chips Family Pack	16 cm / 2	2	\$10.00	0	0.0%

P2 is the clear demand leader: its 506 paid units represent approximately **55% of all paid units**, and its \$1,619.20 revenue represents approximately **57% of paid revenue**. Its 506 consumer TAKE events reconcile exactly to 506 SALE units, with no returns, no misplacement flags, and a strong mean recognition probability of 0.917. This is the strongest available case for additional space.

P4 is the clearest space-efficiency candidate. It occupies **16 cm and two facings** but generated only 2 paid units in 45 days. Its 2 paid units are consistent across transactions and recognition, so the low sales signal is not caused by an obvious reconciliation gap. Reducing it from two facings to one would release 8 cm while retaining a presence in the cooler.

P6 is a serious underperformer by paid demand but has a very different pattern from an ordinary slow seller. It generated only 18 SALE units but 255 ZERO units, meaning 255 pickups were returned without charge. The 255 ZERO units represent approximately **90% of all ZERO units** in the dataset, and its pickup-level return rate is 93.4%. Consumer P6 TAKE events are also lower-confidence than most products: mean probability is 0.863 and 56 events are below 0.80. That combination supports an investigation and likely space reduction, but it does not by itself prove that shoppers dislike Kombucha; it could indicate a recognition, labeling, price, packaging, or placement problem.

Recommendations

1. Widen P2 — Energy Drink B at shelf 1, cell 2

Proposed change: Increase P2 from **one 8 cm facing to two 8 cm facings** at **shelf 1, cell 2**, subject to confirming physical capacity and shelf-cell mechanics.

Basis: P2 sold 506 units and generated \$1,619.20, far ahead of the next product at 107 units. Transactions and recognition reconcile exactly at 506 units. There were no ZERO returns, no misplacement flags, and mean recognition confidence was 0.917. Stocker records show 23 P2 placement events totaling 506 units, consistent with repeated replenishment of a product that is moving substantially faster than every other item.

Implementation note: The added 8 cm could be funded by the proposed reduction of one P4 facing, but the physical layout should be checked because the planogram describes cells and facings separately and does not explicitly state whether adjacent products can share a cell or whether a facing increase requires a new cell assignment.

2. Shrink P4 — Chips Family Pack at shelf 1, cell 4

Proposed change: Reduce P4 from **two facings / 16 cm to one facing / 8 cm** at **shelf 1, cell 4**. Reallocate the released 8 cm to P2 or reserve it for a controlled replacement test.

Basis: P4 produced only 2 paid units and \$10.00 over 45 days while consuming the largest footprint in the planogram: 16 cm. Its 2 SALE units match its 2 consumer TAKE recognition events, and there were no returns or data-quality flags for those events. This is a low-demand, low-ambiguity signal, making a one-facing reduction more defensible than an immediate full removal.

Implementation note: I recommend shrinking rather than removing because the sample records only 2 sales; keeping one facing preserves availability while testing whether the item can remain viable with a smaller footprint.

3. Investigate P6 — Kombucha at shelf 2, cell 2 before a final placement decision

Proposed change: Put P6 through a short operational/data-quality review. If the findings confirm that the events are correctly attributed and the product is genuinely being repeatedly picked up and returned, reduce P6 from **one 8 cm facing to zero or a smaller test allocation** at **shelf 2, cell 2**, and use the space for P2 or a replacement item.

Basis: P6 had only 18 paid units and \$81.00 revenue, but 255 ZERO units. Its 93.4% return rate is an extreme outlier: the next-highest product, P8, was 15.1%. P6 also has 56 consumer events below 0.80 recognition probability, compared with zero such events for P1–P5 and P7–P8 under that threshold in this extract. The direction is therefore unfavorable, but the correct action is conditional because recognition confidence is materially weaker for P6.

Checks before changing the slot: Verify the Kombucha SKU, packaging, and shelf weight profile; review whether the zero-charge records are caused by shoppers changing their minds or by the system failing to maintain product identity; and compare the event timestamps against stocker placements and inventory counts. If those checks pass, P6 is the strongest candidate for removal or replacement after P4.

Data-quality concerns and limitations

The most important issue is **shelf 2, cell 3**, currently mapped to P7 — Mystery Wrap. There are 49 consumer TAKE events attributed to UNKNOWN in that exact slot. Every one is flagged `misplace=True`, and the average recognition probability is only 0.393. At the same time, there are 26 consumer P7 TAKE events at the same slot with mean probability 0.879, and transactions report 26 paid P7 sales. The 26 recognized P7 takes and 26 P7 sales reconcile, but the additional 49 UNKNOWN takes do not have a cataloged product or corresponding UFO transaction rows. This could represent a second product physically entering the slot, a catalog/planogram mapping issue, a sensor interpretation problem, or events that should not be counted as valid shopper takes. I would not widen, remove, or replace P7 until this is explained.

There are **no UFO transaction rows** in the supplied transactions file, despite the existence of UNKNOWN recognition events. This is not necessarily contradictory—the systems may handle unmatched recognition and billing differently—but it means the transaction file cannot be used to quantify the commercial effect of those 49 UNKNOWN events.

The data supports directional decisions but not a precise uplift estimate. The observation period is 45 days, and there is no explicit out-of-stock flag, inventory-on-hand series, margin information, shelf-capacity constraint, or controlled before/after test. In particular, high sales could partly reflect stock availability, and low sales could partly reflect stockouts or replenishment gaps. Therefore, I recommend implementing P2/P4 changes as a monitored test and measuring paid units, return rate, and recognition confidence after the change.

Files produced

File	Contents
planogram_recommendations_report.md	This written report
planogram_investigation.xlsx	Investigation workbook with pivot-style tables and validation tabs
planogram_analysis.py	Reproducible analysis script

focused_checks.py

Focused reconciliation and anomaly checks